Algorithm 1: Stage 1 Multi-Channel Candidate Retrieval & Controlled Negative Sampling

Input: Customer $u \in U$, Transactions D ($t < T_{cutoff}$), Catalog I, Cohorts A, Ratio $\lambda_{neg} = 5$

Output: Balanced training set $T_u = \{(u, i, y)\}$ with $y \in \{0, 1\}$

$C_u \leftarrow \emptyset$

$C_{repurchase} \leftarrow \{i \in I \mid \exists (u, i, t) \in D \wedge t \geq T_{cutoff} - 28d\}$

$C_{bestseller} \leftarrow \text{TopK}(\{i \in I \mid \text{sales_count}(i, [T_{cutoff} - 7d, T_{cutoff}])\}, K = 12)$

$C_{age} \leftarrow \text{TopK}(\{i \in I \mid \text{sales_by_cohort}(i, \text{age_group}(u))\}, K = 8)$

$C_{copurchase} \leftarrow \{j \in I \mid \exists i \in C_{repurchase} \wedge \text{co_purchased}(i, j) \geq 5\}$

$C_u \leftarrow \text{Truncate}(C_{repurchase} \cup C_{bestseller} \cup C_{age} \cup C_{copurchase}, C_{max} = 100)$

$Y_{true} \leftarrow \{i \in I \mid \exists (u, i, t) \in D \wedge t \in [T_{cutoff}, T_{cutoff} + 7d]\}$

$\text{Positives} \leftarrow \{(u, i, 1) \mid i \in (C_u \cap Y_{true})\}$

$N_{cand} \leftarrow C_u \setminus Y_{true}$

$k_{neg} \leftarrow \min(|N_{cand}|, \lambda_{neg} \times |\text{Positives}|)$

$\text{Negatives} \leftarrow \text{SampleWithoutReplacement}(N_{cand}, k_{neg})$

return $T_u \leftarrow \text{Positives} \cup \{(u, j, 0) \mid j \in \text{Negatives}\}$